IT 301C

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DATE:

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APPLICATION LAYER PROTOCOLS

- HYPERTEXT TRANSFER PROTOCOL (HTTP) - TRANSFERS PAGES BETWEEN CLIENT AND SERVER.
- HYPERTEXT TRANSFER PROTOCOL SECURE (HTTPS) - SECURE VERSION OF HTTP USING SSL/TLS ENCRYPTION.
- FILE TRANSFER PROTOCOL (FTP) - TRANSFERS FILES BETWEEN COMPUTERS.
- SECURE FILE TRANSFER PROTOCOL (SFTP) - SECURE FILE TRANSFER OVER SSH.
- SIMPLE MAIL TRANSFER PROTOCOL (SMTP) - SENDS EMAIL MESSAGES.
- POST OFFICE PROTOCOL VERSION 3 (POP3) - RETRIEVES EMAIL FROM A SERVER (DOWNLOADS).
- INTERNET MESSAGE ACCESS PROTOCOL (IMAP) - RETRIEVES AND MANAGES EMAIL DIRECTLY ON THE SERVER.
- DOMAIN NAME SYSTEM (DNS) - TRANSLATES DOMAIN NAMES INTO IP ADDRESSES.
- DYNAMIC HOST CONFIGURATION PROTOCOL (DHCP) - AUTOMATICALLY ASSIGNS IP ADDRESSES TO DEVICES.
- TELECOMMUNICATIONS NETWORK PROTOCOL (TELNET) - REMOTE LOGIN (INSECURE).
- SECURE SHELL (SSH) - SECURE REMOTE LOGIN AND COMMAND EXECUTION.
- SIMPLE NETWORK MANAGEMENT PROTOCOL (SNMP) - MONITORS AND MANAGES NETWORK DEVICES.

NETWORK TIME PROTOCOL (NTP) - SYNCHRONIZES CLOCKS ACROSS DEVICES.

- LIGHTWEIGHT DIRECTORY ACCESS PROTOCOL (LDAP) - ACCESS AND MANAGES DIRECTORY SERVICES.
- SESSION INITIATION PROTOCOL (SIP) - ESTABLISHES AND MANAGES VOIP CALLS.
- REAL-TIME TRANSFER PROTOCOL (RTP) - DELIVERS AUDIO/VIDEO STREAMS IN REAL TIME.

TRANSPORT LAYER PROTOCOLS

- TRANSMISSION CONTROL PROTOCOL (TCP) - RELIABLE, CONNECTION-ORIENTED COMMUNICATION.
- USER DATAGRAM PROTOCOL (UDP) - FAST, CONNECTIONLESS (USED FOR STREAMING/GAMING)

INTERNET LAYER PROTOCOL

- INTERNET PROTOCOL (IPv4/IPv6) - PROVIDES ADDRESSING AND ROUTING OF PACKETS.
- INTERNET CONTROL MESSAGE PROTOCOL (ICMP) - SENDS ERROR MESSAGES AND DIAGNOSTICS (PING/TRACEROUTE)
- ADDRESS RESOLUTION PROTOCOL (ARP) - MAPS IP ADDRESSES TO MAC ADDRESSES.
- REVERSE ADDRESS RESOLUTION PROTOCOL (RARP) - MAPS MAC ADDRESSES BACK TO IP (LEGACY).
- INTERNET PROTOCOL SECURITY (IPSEC) - SECURES IP COMMUNICATIONS WITH AUTHENTICATION AND ENCRYPTION.

DATA LINK LAYER PROTOCOLS

- ETHERNET (IEEE 802.3) - WIRED LAN STANDARD.
- WIRELESS FIDELITY (WIFI/IEEE 802.11) - WIRELESS LAN STANDARD.
- POINT-TO-POINT PROTOCOL (PPP) - USED FOR DIRECT CONNECTIONS BETWEEN TWO NODES.
- HIGH-LEVEL DATA LINK CONTROL (HDLC) - PROVIDES FRAMING AND ERROR DETECTION IN WAN LINKS.

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- FRAME RELAY - LEGACY WAN PROTOCOL FOR PACKET SWITCHING.
- ASYNCHRONOUS TRANSFER MODE (ATM) - HIGH-SPEED CELL-BASED SWITCHING.

ROUTING PROTOCOLS

- ROUTING INFORMATION PROTOCOL (RIP) - DISTANCE-VECTOR ROUTING USING HOP COUNT.
- OPEN SHORTEST PATH FIRST (OSPF) - LINK-STATE ROUTING USING SHORTEST PATH ALGORITHM.
- ENHANCED INTERIOR GATEWAY ROUTING PROTOCOL (EIGRP) - CISCO PROPRIETARY HYBRID ROUTING PROTOCOL.
- BORDER GATEWAY PROTOCOL (BGP) - MANAGES ROUTING BETWEEN AUTONOMOUS SYSTEM (INTERNET BACKBONE).
- INTERMEDIATE SYSTEM TO INTERMEDIATE SYSTEM (IS-IS) - LINK-STATE ROUTING PROTOCOL SIMILAR TO OSPF.

SECURITY AND TUNNELING.

- SECURE SOCKETS LAYER (SSL) - ENCRYPTS COMMUNICATION FOR SECURITY (LEGACY).
- TRANSPORT LAYER SECURITY (TLS) - MODERN ENCRYPTION PROTOCOL REPLACING SSL.
- INTERNET PROTOCOL SECURITY (IPSEC) - SECURES IP COMMUNICATIONS AND VPN TUNNELING.
- GENERIC ROUTING ENCAPSULATION (GRE) - ENCAPSULATES PACKETS FOR TUNNELING.
- POINT-TO-POINT TUNNELING PROTOCOL (PPTP) - VPN TUNNELING PROTOCOL (LEGACY).
- LAYER 2 TUNNELING PROTOCOL (L2TP) - VPN TUNNELING PROTOCOL, OFTEN USED WITH IPSEC.

VICTORY